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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The last element is returned
c. The search is considered unsuccessful ✓
d. The first element is returned

Question **2**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O🤜
c. O(log n)
d. O(1) ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓
b. It always starts from the beginning of the list
c. It works on unsorted lists
d. It compares elements sequentially

Question **4**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Binary search

c. Hash search

d. Linear search ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list

b. The search backtracks to find duplicate elements

c. The search stops immediately ✓

d. The search starts over from the beginning

Question **6**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist

b. In the middle of the list

c. In the right sublist ✓

d. At the beginning of the list

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It is slower than linear search for small lists
- c. It does not work with negative numbers
- d. It can only be applied to large lists

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the entire list
- b. In the right sublist
- c. At the beginning of the list
- d. In the left sublist ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Linear search ✓
- b. Jump search
- c. Binary search
- d. Hashing

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(1)
- b. O👎
- c. O(log n) ✓
- d. O(n log n)

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓

b. Number values should only be present

Number values should only be present

c. The list must be sorted

d. There should be direct access to the middle element in any sublist

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements

b. When the list is unsorted

c. When the list is very small

d. When the list is sorted ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function

b. By comparing each element sequentially

c. By starting from the first element

d. By dividing the list length by two ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search can be applied to both sorted and unsorted lists. ✓
- b. Linear search is more efficient than binary search on large lists.
- c. Linear search divides the list into halves.
- d. Linear search requires the list to be sorted.

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- b. Must use a sorted array
- c. There must be a mechanism to access middle element directly
- d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed